

Mathematics A

Unit 1 • Chapter OL-T21 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 18 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T21 • Expertise Q16+ • 4 items • 18 marks

Chapter	Items	Marks	Starts
01. OL-T21 Algebraic Fractions	4	18	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Fractions

Unit 1 • Expertise • 18 marks • 4 item(s)

November 2023 | 4MA1-2H Q24 | 5 marks

Equation containing algebraic fractions

November 2024 | 4MA1-2H Q23 | 4 marks

Divide factorisable algebraic fractions

January 2021 | 4MA1-2H Q21 | 4 marks

Multiply factorisable algebraic fractions

May-Nov 2020 | 1H Q21 | 5 marks

Substitute a variable relation into a fraction

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q24 whole

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

WORKING SPACE

4MA1-2H Q24 whole

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x = \dots\dots\dots$

Solve a rational algebraic equation

5 marks

whole – Solve a rational algebraic equation

Combine the bracketed fractions

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x+3x-4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}.$$

Factor the first fraction

$$45x^3 - 80x = 5x(3x-4)(3x+4), \quad 3x^2 + x - 4 = (3x+4)(x-1).$$

Cancel common factors

The left-hand side simplifies to 20 (with the original denominator restrictions retained).

Solve the remaining equation

$$20 = \frac{4(x+2)}{5x-8} \Rightarrow 20(5x-8) = 4(x+2).$$

Find and check x

$$96x = 168 \Rightarrow x = \frac{7}{4}, \text{ which is not an excluded denominator value.}$$

Final answer

$$x = 7/4$$

4MA1-2H Q23 M01

4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

WORKING SPACE

Simplify an algebraic fraction**4 marks****M01 – Simplify an algebraic fraction****Factorise the first numerator**

$$16x^2 - 36 = 4(2x - 3)(2x + 3).$$

Factorise the remaining terms

$$2x^2 + 7x + 6 = (2x + 3)(x + 2), \quad x^2 - 5x - 14 = (x - 7)(x + 2).$$

Cancel

The quotient simplifies to $4(2x-3)$.

Subtract the linear expression

$$4(2x-3)-(7+8x)=-19, \text{ so } n=-19.$$

Final answer

$$n = -19$$

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

Worked solution

January 2021 | Question 21 | 4 marks

Family: Multiply or divide algebraic fractions • Variant: Multiply factorisable algebraic fractions

Method / working

1. Factor $25x^2-64=(5x-8)(5x+8)$, $x^2-8x+15=(x-5)(x-3)$, and $5x^2-13x-6=(5x+2)(x-3)$.
2. Factor and cancel common factors before combining.
3. Put the remaining terms over a common denominator.
4. The simplified single fraction is $54/(5x+2)$.

Final answer: $54/(5x+2)$

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

Answer reference | May-Nov 2020 | Question 21

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

21	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for either $(3x+2)(3x-2)$ or $(3x+2)(x-5)$	M2 for $\frac{9x^2 - 4}{(9x^2 - 4)(x-5)} =$ $\frac{1}{(x-5)}$
	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for $(3x+2)(3x-2)$ and $(3x+2)(x-5)$	
	E.g. of denominators $(3x-2)(3x^2-13x-10)(x-1)$ or $(3x-2)(3x+2)(x-5)(x-1)$ or $9x^4 - 54x^3 + 41x^2 + 24x - 20$ or $(3x+2)(x-5)(x-1)$ or $3x^3 - 16x^2 + 3x + 10$ or $(3x-2)(x-5)(x-1)$ or $3x^3 - 20x^2 + 27x - 10$ or $(x-5)(x-1)$ or $x^2 - 6x + 5$			M1 (indep) ft their fractions for use of a correct common denominator for 2 fractions with algebraic denominators NB: fractions need not be simplified	
	$\frac{x-1-7(x-5)}{(x-5)(x-1)}$ or $\frac{x-1-7x+35}{(x-5)(x-1)}$ or $\frac{x-1-7(x-5)}{x^2-6x+5}$ or $\frac{x-1-7x+35}{x^2-6x+5}$ oe			M1 for a correct fraction with a correct quadratic denominator – may or may not be expanded which leads to a correct answer	
		$\frac{2(17-3x)}{(x-5)(x-1)}$	5	A1 accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct	
Total 5 marks					

Worked solution

May-Nov 2020 | Question 21 | 5 marks

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

Method

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10}$$

$$-\frac{7}{x-1}$$

Factor:

$$9x^2-4=(3x-2)(3x+2)$$

$$3x^2-13x-10=(3x+2)(x-5)$$

So the first product becomes

$$\frac{1}{3x-2} \times$$

$$\frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

Cancel common factors:

$$\frac{1}{x-5}$$

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

Use the common denominator $(x-5)(x-1)$:

$$\frac{(x-1)-7(x-5)}{(x-5)(x-1)}$$

$$= \frac{(x-1-7x+35)}{(x-5)(x-1)}$$

$$= \frac{(34-6x)}{(x-5)(x-1)}$$

Answer: $\boxed{\frac{(34-6x)}{(x-5)(x-1)}}$